

LAB 2: AGILE BACKLOG CREATION & SPRINT SIMULATION IN JIRA

Urban Rooftop Gardening Co-op App

Software Engineering Laboratory Submission

Project: Urban Rooftop Gardening Co-op App
Jira project: SP

Objective

Use Jira to convert functional requirements into Agile backlog items, prioritize and estimate them, simulate sprints, and analyse progress.

1. Epics and Backlog Structure

The project was organized into three major functional themes. Each Epic was broken down into related work items covering rooftop plot and crop management, soil knowledge sharing, and surplus produce trading.

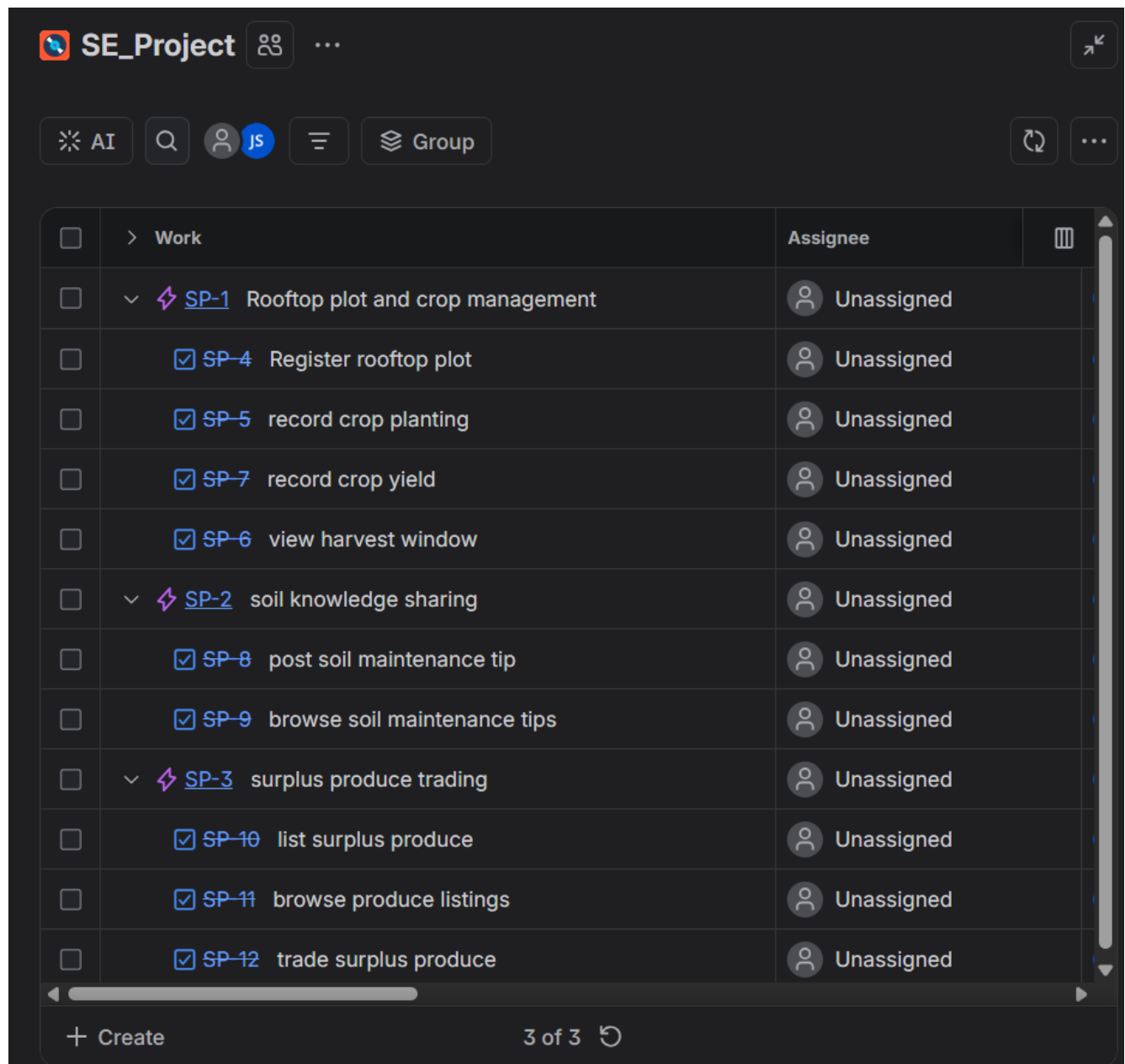


Figure 1. Jira List view showing the three Epics and their child work items.

Epic 1: Rooftop plot and crop management

Epic 2: Soil knowledge sharing

Epic 3: Surplus produce trading

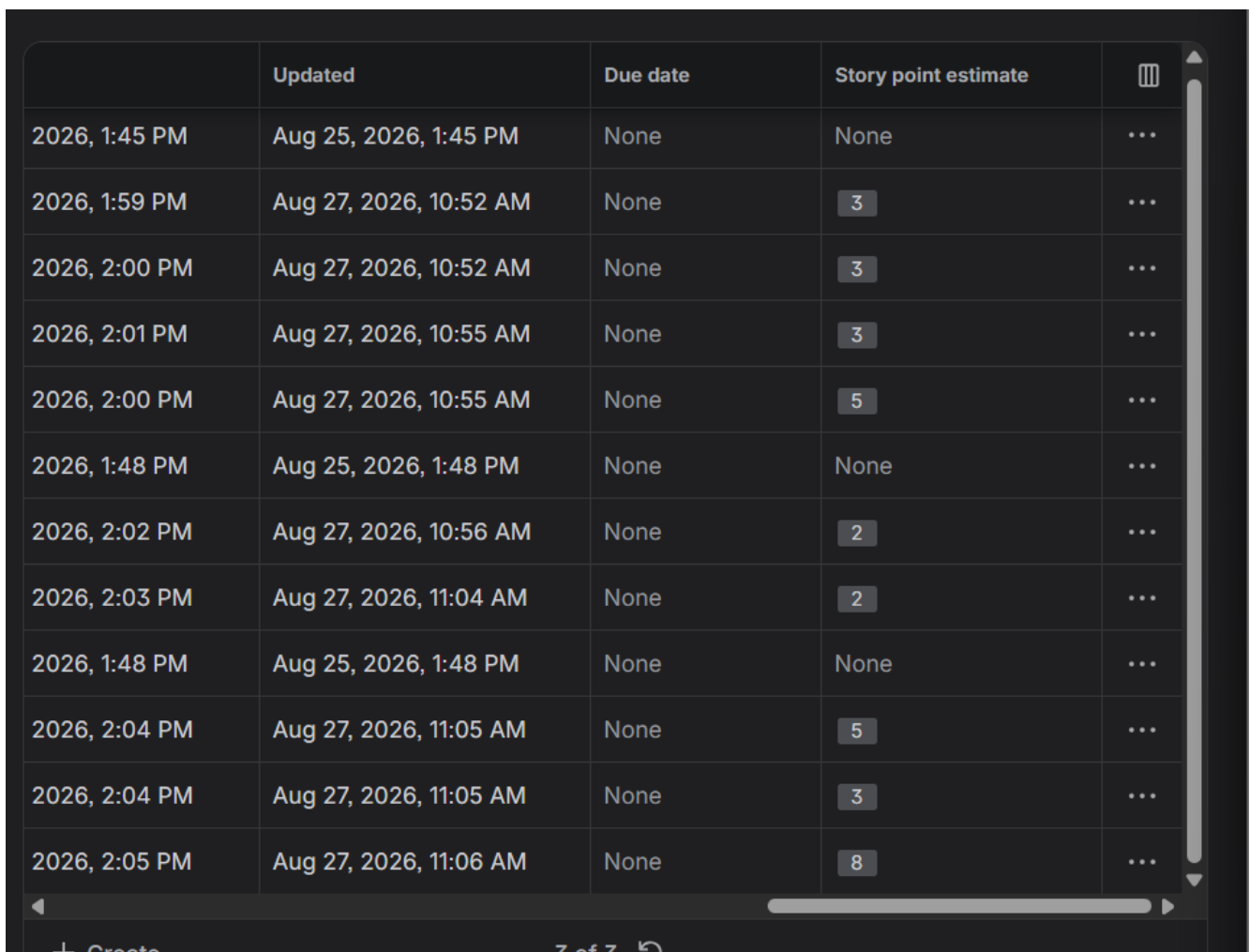
The handout requires Epics to be created first and then populated with related User Stories/work items.

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2. Priorities and Story Point Estimation

The work items were prioritized according to user impact and estimated using Fibonacci-style story points, as required by the lab.

Work item	Points
Register rooftop plot	3
Record crop planting	3
View harvest window	5
Record crop yield	3
Post soil maintenance tip	2
Browse soil maintenance tips	2
List surplus produce	5
Browse produce listings	3
Trade surplus produce	8

A screenshot of a Jira Agile Backlog interface. The table displays a list of tasks with their respective story point estimates. The 'Story point estimate' column shows values of 3, 3, 3, 5, 2, 2, 5, 3, and 8 for various tasks. The interface includes a dark theme, a sidebar on the left, and a bottom bar with a '+ Create' button and a '3 of 3' indicator.

	Updated	Due date	Story point estimate	
2026, 1:45 PM	Aug 25, 2026, 1:45 PM	None	None	...
2026, 1:59 PM	Aug 27, 2026, 10:52 AM	None	3	...
2026, 2:00 PM	Aug 27, 2026, 10:52 AM	None	3	...
2026, 2:01 PM	Aug 27, 2026, 10:55 AM	None	3	...
2026, 2:00 PM	Aug 27, 2026, 10:55 AM	None	5	...
2026, 1:48 PM	Aug 25, 2026, 1:48 PM	None	None	...
2026, 2:02 PM	Aug 27, 2026, 10:56 AM	None	2	...
2026, 2:03 PM	Aug 27, 2026, 11:04 AM	None	2	...
2026, 1:48 PM	Aug 25, 2026, 1:48 PM	None	None	...
2026, 2:04 PM	Aug 27, 2026, 11:05 AM	None	5	...
2026, 2:04 PM	Aug 27, 2026, 11:05 AM	None	3	...
2026, 2:05 PM	Aug 27, 2026, 11:06 AM	None	8	...

Figure 2. Jira Story point estimate column showing the assigned Fibonacci-style estimates.

Total estimated effort: **34 story points**.

The lab states that story points represent relative effort, complexity, time/risk and uncertainty, and recommends Fibonacci values for estimation. ■filecite■turn10file0■L153-L185■

3. Sprint 1 Simulation

Sprint 1 was configured as a one-week sprint. All nine work items were initially added to the sprint. During the simulation, work was moved through To Do, In Progress and Done to represent progress.

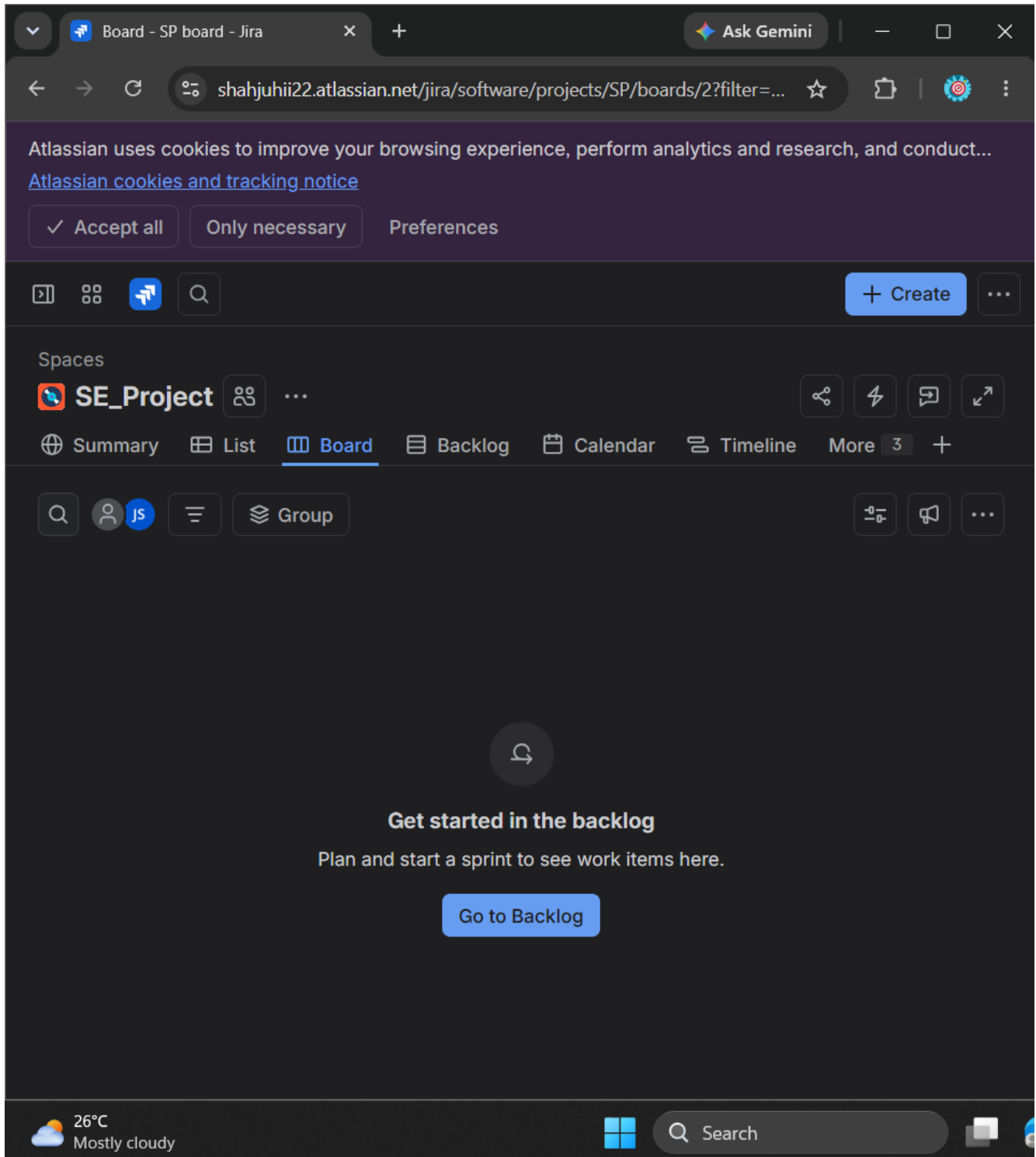


Figure 3. Sprint 1 Active Sprint board after simulated progress.

Sprint 1 result: **5 work items completed**, representing **16 story points**, with **18 story points remaining**.

The handout requires a one-week Sprint 1, selected stories to be dragged into the sprint, and tasks to be progressed from To Do to In Progress to Done. ■filecite■turn10file0■L206-L230■

4. Sprint 2 Simulation

The four open work items from Sprint 1 were moved into Sprint 2. Sprint 2 was also configured for one week, and all four remaining items were progressed to Done.

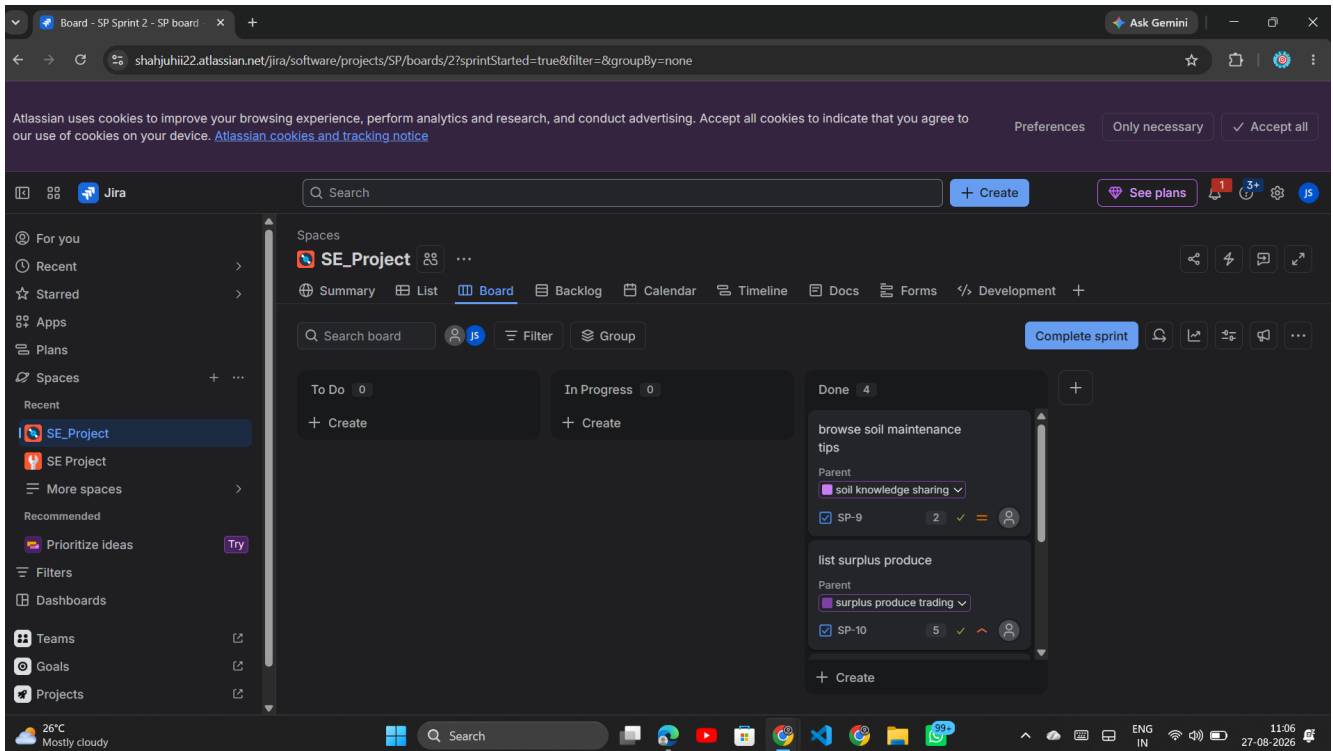


Figure 4. Sprint 2 board showing all four remaining work items completed.

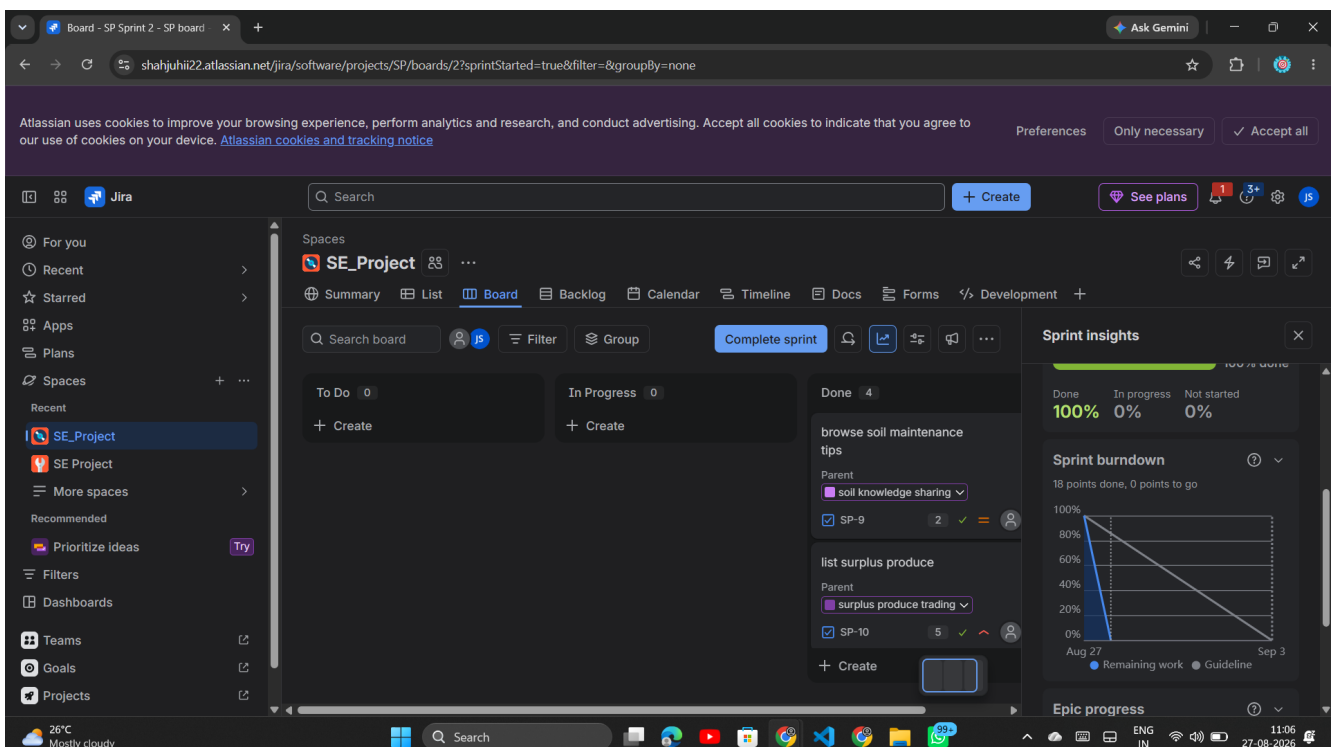


Figure 5. Jira Sprint 2 completion summary showing 100% completion.

Sprint 2 result: 4 work items completed, representing the remaining 18 story points.

The lab explicitly requires students to finish two sprints. ■filecite■turn10file0■L229-L237■

5. Burndown Chart

The Jira Burndown Chart was used to observe completed and remaining work during the sprint. At the captured point in Sprint 1, the chart showed 16 points completed and 18 points remaining from the 34-point plan.

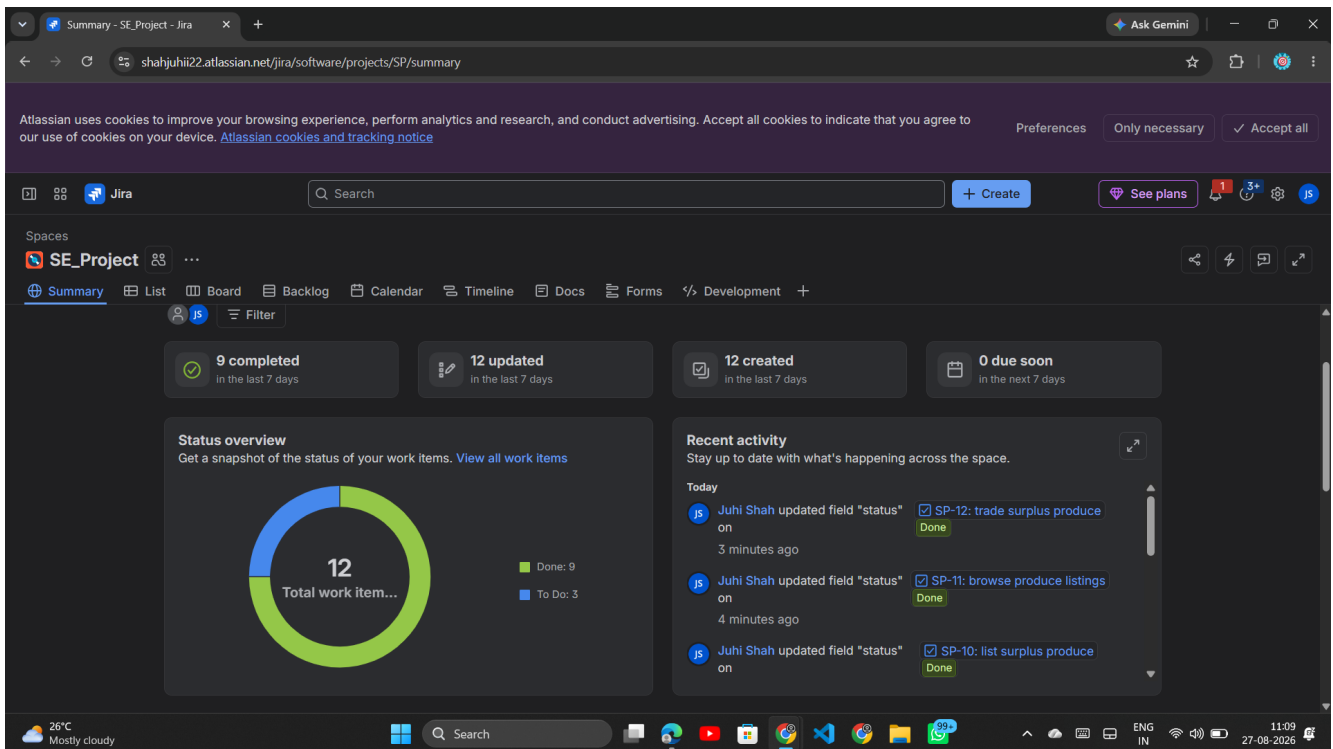


Figure 6. Jira Burndown Chart for Sprint 1.

The handout explains that a burndown chart represents the total work remaining and helps assess whether the team is on track against the available sprint time. ■filecite■turn10file0■L238-L265■

6. Reflection

1. Did your estimations reflect the actual effort?

The estimates were reasonably representative for the simulated work. The smaller tasks were assigned 2 or 3 points, while tasks involving more functionality or uncertainty received 5 or 8 points. The first sprint completed 16 of the planned 34 points, while the remaining 18 points were completed in Sprint 2. This suggests that the estimates provided a useful relative measure of effort, although the simulation was not a real development implementation.

2. Was your backlog well-prioritized?

Yes. Higher-impact core functionality was given higher priority, while supporting knowledge-sharing work was placed at a lower priority. This created a logical order for implementing the rooftop gardening and produce-trading features.

3. How did your simulated sprint align with your plan?

The sprint plan was followed by selecting all nine work items for Sprint 1, progressing them through the Scrum board, completing Sprint 1, and carrying the four unfinished items into Sprint 2. Sprint 2 then completed the remaining 18 story points. Overall, the planned 34 story points were completed across the two sprints.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart showed that 16 story points had been completed while 18 remained at the captured point in Sprint 1. This indicates that the team had completed nearly half of the planned effort but still had substantial work remaining. It also demonstrates how the chart can be used to compare actual progress with the expected trend and judge whether the remaining workload is achievable within the sprint.

7. Conclusion

The Jira exercise converted the Urban Rooftop Gardening Co-op App requirements into an Agile backlog, organized the work into three Epics, assigned priorities and story points, and simulated two Scrum sprints. Sprint 1 completed 16 story points and Sprint 2 completed the remaining 18 points, resulting in 34 story points completed overall. The Burndown Chart provided a visual measure of progress and remaining work.

Submission evidence included:

- Jira backlog with Epics and child work items
- Story point assignments
- Sprint 1 Active Sprint board
- Sprint 2 completion evidence
- Burndown Chart
- Reflection answers